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Practical No 3

Course

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3.1.1. Numpy Array Operations

Write a Python program to create a NumPy array based on user input and display the following:

- The created NumPy array.
- The number of dimensions of the array using `ndim`.
- The shape of the array using `shape`.
- The total number of elements in the array using `size`.

Assume all input elements are valid integers.

Input Format:

- The first line contains two space-separated integers, `r` and `c`, representing the number of rows and the number of columns.
- The next `r` lines contain `c` space-separated integers representing the elements of the array, entered row by row.

Output Format:

- The created NumPy array (printed as a 2D array).
- An integer representing the number of dimensions of the array.
- A tuple representing the shape of the array.
- An integer representing the total number of elements in the array.

Note: Use `reshape()` function to reshape the input array with the specified number of rows and columns.

Sample Test Cases

```
1 # import numpy as np
2
3 # write your code here...
4 import numpy as np
5
6 rows, cols = map(int, input().split())
7
8 elements = []
9
10 for _ in range(rows):
11     elements.extend(map(int, input().split()))
```

Average time: 0.011 s, Maximum time: 0.021 s

2 out of 2 shown test case(s) passed
10 out of 10 hidden test case(s) passed

Test case 1: 21 ms

Expected output	Actual output
3 4	3 4
1 2 3 4	1 2 3 4
5 6 7 8	5 6 7 8
9 10 11 12	9 10 11 12
[[1 2 3 4]	[[1 2 3 4]
[5 6 7 8]]	[5 6 7 8]]

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3.2.1. Numpy: Matrix Operations

The given code takes two 3×3 matrices, `matrix_a`, and `matrix_b`, as input from the user and converts them into NumPy arrays.

Task:

You are required to compute and display the results of the following matrix operations:

- Addition (`matrix_a + matrix_b`)
- Subtraction (`matrix_a - matrix_b`)
- Element-wise Multiplication (`matrix_a * matrix_b`)
- Matrix Multiplication (`matrix_a · matrix_b`)
- Transpose of Matrix A

Input Format:

- The user will input 3 rows for `matrix_a`, each containing 3 integers separated by spaces.
- Similarly, the user will input 3 rows for `matrix_b`, each containing 3 integers separated by spaces.

Output Format:

The program should display the results of the operations in the following order:

- The result of Addition.
- The result of Subtraction.
- The result of Element-wise Multiplication.
- The result of Matrix Multiplication.
- The Transpose of Matrix A.

```
1 # import numpy as np
2
3 ## Input matrices
4 # print("Enter Matrix A:")
5 # matrix_a = np.array([list(map(int, input().split())) for i in range(3)])
6
7 # print("Enter Matrix B:")
8 # matrix_b = np.array([list(map(int, input().split())) for i in range(3)])
9
```

Average time: 0.018 s, Maximum time: 0.024 s

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1: 22 ms

Expected output	Actual output
Enter Matrix A:	Enter Matrix A:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Matrix B:	Enter Matrix B:
1 1 1	1 1 1

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3.2.2. Numpy: Horizontal and Vertical Stacking of Arrays

02:30

You are given two arrays `arr1` and `arr2`. You need to perform horizontal and vertical stacking operations on them using NumPy.

- Horizontal Stacking:** Stack the two matrices horizontally (side by side).
- Vertical Stacking:** Stack the two matrices vertically (one below the other).

Input Format:

- The program should first prompt the user to input two 3x3 arrays.
- Each array consists of 3 rows, and each row contains 3 space-separated integers.
- The user will input the two arrays row by row.

Output Format:

- The program should display the result of the Horizontal Stack (side-by-side stacking) of the two arrays.
- The program should then display the result of the Vertical Stack (one below the other) of the two arrays.

Sample Test Cases

stacking.py

Submit

Debugger

```
1 import numpy as np
2
3 # Input matrices
4 print("Enter Array1:")
5 arr1 = np.array([list(map(int, input().split())) for i in range(3)])
6
7 print("Enter Array2:")
8 arr2 = np.array([list(map(int, input().split())) for i in range(3)])
9
10 # Perform horizontal stacking (hstack)
11 horizontal=np.hstack((arr1,arr2))
```

Average time 0.014 s 14.00 ms Maximum time 0.015 s 15.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 15 ms Debug

Expected output	Actual output
Enter Array1:	Enter Array1:
1 2 3	1 2 3
4 5 6	4 5 6
7 8 9	7 8 9
Enter Array2:	Enter Array2:
4 5 6	4 5 6

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3.2.3. Numpy: Custom Sequence Generation

02:30

Write a Python program that takes the following inputs from the user:

- Start value: The starting point of the sequence.
- Stop value: The sequence should end before this value.
- Step value: The increment between each number in the sequence.

The program should then generate a sequence using `numpy` based on these inputs and print the generated sequence.

Input Format:

- The user will input three integer values: start, stop, and step, each on a new line.

Output Format:

- The program should print the generated sequence based on the input values.

Sample Test Cases

customS...

Submit

Debugger

```
1 import numpy as np
2
3 # Take user input for the start, stop, and step of the sequence
4 start = int(input())
5 stop = int(input())
6 step = int(input())
7
8 # Generate the sequence using np.arange()
9 seq=np.arange(start,stop,step)
10 # Print the generated sequence
11 print(seq)
```

Average time 0.007 s 6.75 ms Maximum time 0.010 s 10.00 ms

2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed

Test case 1 10 ms Debug

Expected output	Actual output
3	3
10	10
2	2
[3 5 7 9]	[3 5 7 9]

Test case 2 7 ms

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3.2.4. Numpy: Arithmetic and Statistical Operations, Mathematical Operations, Bitwise Operations

You are given two arrays A and B. Your task is to complete the function `array_operations`, which will convert these lists into NumPy arrays and perform the following operations:

- Arithmetic Operations:**
 - Compute the element-wise sum, difference, and product of the two arrays.
- Statistical Operations:**
 - Calculate the mean, median, and standard deviation of array A.
- Bitwise Operations:**
 - Perform bitwise AND, bitwise OR, and bitwise XOR on the arrays (ex: A_1 OR B_1).

Input Format:

- The first line contains space-separated integers representing the elements of array A.
- The second line contains space-separated integers representing the elements of array B.

Output Format:

- For each operation (arithmetic, statistical, and bitwise), print the results in the specified format as shown in sample test cases.

Sample Test Cases

different...

```
1 import numpy as np
2
3 def array_operations(A, B):
4
5     # Convert A and B to NumPy arrays
6     A = np.array(A)
7     B = np.array(B)
8
9     # Arithmetic Operations
10    sum_result = A+B
11    diff_result = A-B
12    prod_result = A*B
```

Average time0.008 s8.00 msMaximum time0.011 s11.00 ms

1 out of 1 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 111 msDebug

Expected output

1 2 3 45 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

Actual output

1 2 3 45 6 7 8

Element-wise Sum: 6 8 10 12

Element-wise Difference: -4 -4 -4 -4

Element-wise Product: 5 12 21 32

Mean of A: 2.5

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3.2.5. Numpy: Copying and Viewing Arrays

The given code takes a list of integers as input and converts it into a NumPy array. Your task is to complete the code by:

- Creating a view of the `original_array` and assigning it to `view_array`.
- Creating a copy of the `original_array` and assigning it to `copy_array`.

After completing these steps, observe how modifying the view affects the `original_array`, while modifying the copy does not.

Input Format:

- A single line of space-separated integers.

Output Format:

- After modifying the view:

Original array after modifying view: <original_array>
View array: <view_array>

- After modifying the copy:

Original array after modifying copy: <original_array>
Copy array: <copy_array>

Sample Test Cases

copyAnd...

```
1 import numpy as np
2
3 inputlist = list(map(int,input().split(" ")))
4
5 # Original array
6 original_array = np.array(inputlist)
7
8 # Create a view
9 view_array = original_array.view()
10
11 # Create a copy
```

Average time0.004 s4.50 msMaximum time0.007 s7.00 ms

2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 17 msDebug

Expected output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

Actual output

10 20 30 40 50 60 70 80

Original array after modifying view: [99 20 30 40 50 60 70 80]

View array: [99 20 30 40 50 60 70 80]

Original array after modifying copy: [99 20 30 40 50 60 70 80]

Copy array: [10 88 30 40 50 60 70 80]

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3.2.6. Numpy: Searching, Sorting, Counting, Broadcasting

05:23

The given code in the editor takes a single array, array1, as space-separated integers as input from the user.

Additionally, it takes the following inputs:

- search_value: The value to search for in the array.
- count_value: The value to count its occurrences in the array.
- broadcast_value: The value to add for broadcasting across the array.

You need to complete the code to perform the following operations:

1. **Searching:** Find the indices where search_value appears in array1 and print these indices.
2. **Counting:** Count how many times count_value appears in array1 and print the count.
3. **Broadcasting:** Add broadcast_value to each element of array1 using broadcasting, and print the resulting array.
4. **Sorting:** Sort array1 in ascending order and print the sorted array.

Input Format:

1. A single line containing space-separated integers representing array1.
2. An integer search_value represents the value to search for in the array.
3. An integer count_value represents the value to count in the array.
4. An integer broadcast_value represents the value to add to each element of the array.

Output Format:

1. The indices where search_value occurs in array1.
2. The count of occurrences of count_value in array1.
3. The array after adding the broadcast_value to each element.

arrayOpe...

Submit

Debugger

```
1 import numpy as np
2
3 # Input array from the user
4 array1 = np.array(list(map(int, input().split()))))
5
6 # Searching
7 search_value = int(input("Value to search: "))
8 count_value = int(input("Value to count: "))
9 broadcast_value = int(input("Value to add: "))
10
11 # Find indices where value matches in array1
```

Average time

Maximum time

0.011 s

0.012 s

10.50 ms

12.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 1

1/2 ms

Debug

Expected output

Actual output

1 1 2 2 2

1 1 2 2 2

Value to search: 1

Value to search: 1

Value to count: 2

Value to count: 2

Value to add: 2

Value to add: 2

[0 1 2]

[0 1 2]

3

3

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3.2.7. Student Data Analysis and Operations

10:50

Write a Python program that takes the file name of a CSV file containing student details, including roll numbers and their marks in three subjects as input, reads the data, and performs the following operations:

- **Print all student details:** Display the complete details of all students, including roll numbers and marks for all subjects.
- **Find total students:** Determine the total number of students in the dataset.
- **Print all student roll numbers:** Extract and print the roll numbers of all students.
- **Print Subject 1 marks:** Extract and print the marks of all students in Subject 1.
- **Find minimum marks in Subject 2:** Identify the lowest marks in Subject 2.
- **Find maximum marks in Subject 3:** Identify the highest marks in Subject 3.
- **Print all subject marks:** Display the marks of all students for each subject.
- **Find total marks of students:** Compute the total marks for each student across all subjects.
- **Find the average marks of each student:** Compute the average marks for each student.
- **Find average marks of each subject:** Compute the average marks for all students in each subject.
- **Find average marks of Subject 1 and Subject 2:** Compute the average marks for Subject 1 and Subject 2.
- **Find average marks of Subject 1 and Subject 3:** Compute the average marks for Subject 1 and Subject 3.
- **Find the roll number of the student with maximum marks in Subject 3:** Identify the student with the highest marks in Subject 3 and print their roll number.
- **Find the roll number of the student with minimum marks in Subject 2:** Identify the student with the lowest marks in Subject 2 and print their roll number.
- **Find the roll number of students who scored 24 marks in Subject 2:** Identify students who obtained exactly 24 marks in Subject 2 and print their roll numbers.
- **Find the count of students who not less than 40 marks in Subject 1:** Count the number of

Operatio...

Submit

Debugger

```
1 import numpy as np
2
3 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
4
5 import numpy as np
6
7 a = np.loadtxt("Sample.csv", delimiter=',', skiprows=1)
8
9 # 1. Print all student details
10 print("All student Details:\n",a)
11
```

Average time

Maximum time

0.013 s

0.013 s

13.00 ms

13.00 ms

1 out of 1 shown test case(s) passed

Test case 1

1/3 ms

Debug

Expected output

Actual output

All student Details:

All student Details:

[[301...67...77...88...]

[[301...67...77...88...]

[302...78...88...77...]

[302...78...88...77...]

[303...45...56...89...]

[303...45...56...89...]

[304...88...98...45...]

[304...88...98...45...]

[305...78...88...99...]

[305...78...88...99...]

306 68 79 90 1

306 68 79 90 1

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